

# Beating the Style Detector: Three Hours of Agentic Research on the AI-Text Arms Race

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Withheld for blind review

**Abstract.** Reproducing an empirical NLP study used to take weeks. Given the released data and a modern agentic-research harness, we redo every experiment of a recent ACL 2026 study on personal-style post-editing of LLM drafts—and add three new ones—with the human investigator acting only as a reviewer-in-the-loop. We reproduce all seven preregistered hypotheses and recover the paper’s headline correlation between perceived self-similarity and embedding-measured self-similarity to three decimal places ( $r=+0.244$ ,  $p<10^{-8}$ ,  $n=648$ ). Under a leakage-free held-out protocol, GPT-5.5 and ClaudeOpus 4.7 close 71–75 % of the style gap to the same-author ceiling on 324 paired tasks, against 24 % for the human post-edit, and beat the human post-edit on  $\sim 80$  % of tasks. We then frame the same data as an AI-text detection arms race. A leave-authors-out linear SVM on LUAR-MUD embeddings reaches AUC 0.93–1.00 across approaches; six diagnostics show that GPT-5.5 detection is mostly a length confound while Opus detection is a genuine stylistic signature. Given  $T=20$  feedback iterations against the frozen detector, an Opus agent flips two of five held-out test mimics to the human half-space and shrinks every margin by an order of magnitude. With moderate effort against a known detector, a frontier LLM can already efficiently lower its own AI-detection probability. All code, 648 mimic drafts, trained detectors, diagnostics, and adversarial trajectories are released.

**Keywords:** Reproducible research · AI-text detection · Agentic research · Authorship style embeddings.

## 1 Introduction

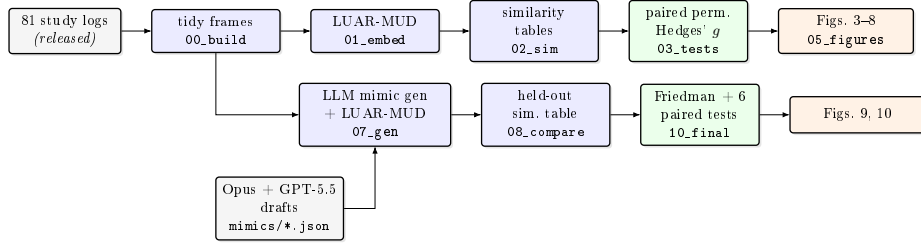
Detecting whether a piece of text was written by a person or by a language model is, fundamentally, an arms race. Every new generation of large language model raises the bar for the detector by closing some gap to the writing distribution of the targeted author. Every new detection approach in turn raises the bar for the generator. The ACL 2026 paper of Baumler et al. [2], which appeared on arXiv on April 27, 2026, makes this concrete: the authors ask whether participants can *post-edit* an LLM-generated draft so that it sounds like them, with stylistic similarity measured by LUAR-MUD [10] and the original draft generated by

OpenAI’s o4-mini. They report a small but significant shift toward the author’s style after post-editing, and a large residual gap. Their study ships the full *data* (81 JSON logs, 486 task observations) but no analysis code, which makes it a clean target for a Track2 RRPR companion paper.

This paper is that companion. The methodological point is straightforward: when released data already exist and an agentic-research harness is available, full reproduction plus three extension experiments can be executed in roughly three hours of real elapsed time rather than weeks. In the present project, the harness, in the spirit of Karpathy’s autoresearch [6] and Zaiss et al. [12], served as the execution layer for the full pipeline: re-implementing the released analysis, generating the LLM-mimic drafts via parallel sub-agents, instantiating the held-out protocol, identifying and correcting an initially shipped leakage bug, and maintaining the manuscript against an explicit requirements checklist. The human investigator remained at the level of methodological supervision, issuing requests such as “check that there is no train/test leakage”, while the harness translated those requests into concrete code changes and re-ran the pipeline. This remains within the RRPR Track2 scope: the scientific object is the companion reproducibility study, and the agentic loop is the means by which it is carried out.

The paper is organised as four contributions. Sections 3–4 re-implement the source paper’s full statistical pipeline from the released logs and reproduce every preregistered hypothesis. The pipeline combines three components: LUAR-MUD [10] as a frozen authorship-style embedding model, paired permutation tests with Hedges’  $g$  [5] as the group-comparison instrument, and *repeated-measures correlation* (rmcorr, Bakdash and Marusich [1]) for the within-subject correlation between perceived and embedding-measured self-similarity. Section 5 extends the design with a leakage-free held-out evaluation that shows one of each participant’s two unassisted controls to a frontier LLM as a style demonstration and reserves the other as the shared evaluation target across all four conditions (o4-mini draft, human post-edit, GPT-5.5 mimic, Opus 4.7 mimic). Section 6 reports a Friedman omnibus, all six paired tests with BH-FDR, and a per-task win-rate against the human-edit threshold. Section 7 re-casts the same data as an AI-text detection task, runs six robustness diagnostics, and closes the loop with an adversarial rewriting experiment in which an Opus 4.7 agent receives the detector’s score on its current draft and is asked to defeat it.

We frame the LLM-mimic prompt explicitly as a *hybrid / known-operator* approach, in the sense of Maier et al. [8,7]: the prompt encodes a known stylistic constraint (the demonstrated style sample), while the LLM provides the unstructured generation step. The same metric-guided idea also drives the adversarial rewriting loop: a frozen detector provides an external objective signal while the frontier model searches over rewrites. In the present paper this signal can only act at inference time, because the frontier agentic models we use do not expose trainable weights; the same principle should be even stronger when the loss is available directly during fine-tuning or pretraining, as in Karpathy’s autoresearch [6]. We therefore use the same template in a narrower RRPR Track2



**Fig. 1:** Pipeline used by the companion code. Solid arrows: data flow. Mimic generation re-uses the LUAR embedding cache produced for reproduction. Stage names match the script files (00\_..., 01\_..., etc.) in the released repository.

sense: a tightly-defined task, a fixed automatic metric, and an auditable companion pipeline [6,12].

All figures, tables, statistical tests, trained detectors, and 648 LLM-mimic drafts can be regenerated by running `make all && make mimic-compare && make final-assessment` and `python3 scripts/11_detection_experiment.py` after `pip install -r requirements.txt`.<sup>1</sup>

## 2 Background

*The source paper.* Baumler et al. [2] run a pre-registered online study with 81 participants ( $n=486$  task observations). Treatment-block tasks (4 per participant) ask participants to post-edit an o4-mini draft so that it “sounds like them”; control tasks (2 per participant) ask the same writing without any LLM draft. Stylistic similarity is measured by cosine similarity in LUAR-MUD [10], a contrastive author-embedding model trained on Reddit. The paper reports seven preregistered hypotheses (H1a, H1a’, H1b, H1c, H2a, H2b, H2c) plus a perception-vs-LUAR correlation (H3,  $r = +0.244 \pm 0.076$ ,  $p < .0001$ ). All seven survive BH-FDR at  $q=0.05$ .

*LUAR-MUD as a known operator.* LUAR-MUD [10] is a frozen, publicly available embedding model that maps a text to a 512-dimensional vector trained so that texts by the same Reddit author have higher cosine similarity than texts by different authors. We treat it as a fixed *known operator* for the pattern-recognition reader: our pipeline does not fine-tune it, the revision SHA is pinned, and every similarity is a deterministic function of the text plus that pinned SHA. This is exactly the framing of hybrid models that Maier et al. [8,7] advocate—the network is fixed, the task lives outside the network (here, the prompt design and the held-out protocol). Treating LUAR this way is what makes the comparison reproducible in Section 5.

<sup>1</sup> A link to the full source repository will be released with acceptance of the paper.

*Comparable LLM-vs-human benchmarks.* The closest setups in the literature on agentic LLM evaluation are Karpathy’s autoresearch [6], where an agent modifies a small GPT-style training stack and keeps changes that lower validation bits per byte under a fixed 5-minute training budget, and Zaiss et al. [12], where agents iteratively design MRI sequences and reconstructions against an automatic image-quality objective. Our setting is narrower but structurally similar: the agent operates against a fixed automatic metric on a constrained task, which is what makes the comparison to a human baseline informative. Because we do not control frontier-model weights, our loop can only optimize inference-time behaviour.

### 3 Reproduction methodology

We obtain the 81 JSON logs from the upstream release. Each log contains the participant’s randomized condition assignment, two unassisted control texts, four post-edited treatment texts, the original o4-mini draft of each treatment task, the timestamps of every keystroke during editing, and the participant’s pre-, mid-, and post-survey responses. We tidy these into a long-format  $486 \times n$  DataFrame (one row per task observation), embed every text with LUAR-MUD pinned to revision 9204529, and persist the embeddings as a single `.npz` cache so the rest of the pipeline is CPU-only. The full pipeline is implemented in Python with standard scientific libraries and Pingouin. Pipeline overview in Fig. 1; for a general accessible introduction to embedding-based pipelines of this kind in adjacent domains see Maier et al. [9].

*Statistical protocol.* Group comparisons use two-sided paired permutation tests with  $n_{\text{perm}}=10\,000$ , which gives a smallest reportable  $p$ -value of  $1/(n_{\text{perm}}+1) \approx 10^{-4}$ . Effect sizes are Hedges’  $g$  [5] with the small-sample correction; 95% CIs come from 1 000 bootstrap resamples. Multiple-testing correction uses Benjamini–Hochberg [3] at  $q=0.05$  over the seven preregistered hypotheses. The H3 correlation between perceived self-similarity (averaged from two Likert items per text) and LUAR self-similarity is fit with repeated-measures correlation [1] as implemented in Pingouin.

*Determinism.* All NumPy and PyTorch random number generators (RNGs) are seeded at process start. Permutation tests draw their own seeded `Generator` per test. Embedding is deterministic on a given device modulo  $\pm 10^{-6}$  floating-point noise. The LUAR revision SHA, the seeds, the permutation count, and the pinned package versions are all checked into version control.

### 4 Reproduction results

Table 1 compares our results, computed entirely from the released logs, against the values printed in Baumler et al. [2]; Fig. 2 re-creates three of the source paper’s published figures from the same logs. All seven preregistered hypotheses

**Table 1:** Reproduction of the seven preregistered hypotheses (H1a... H2c) and the H3 perception-vs-LUAR correlation. PAPER  $g$  is the value printed in Bauml et al. [2]; OURS  $g$  is the value our re-implementation produces from the released logs. ✓ in BH-FDR means the test survives Benjamini–Hochberg correction at  $q=0.05$  over the seven preregistered tests. The H3 row reports the `rmcorr` [1] correlation rather than  $g$ .

| Hyp. | Paper $g$ | Ours $g$ | 95 % CI (ours) | BH-FDR |
|------|-----------|----------|----------------|--------|
| H1a  | +0.55     | +0.52    | [+0.45, +0.61] | ✓      |
| H1b  | −0.41     | −0.63    | [−0.71, −0.55] | ✓      |
| H1a′ | −0.56     | −1.17    | [−1.32, −1.04] | ✓      |
| H1c  | −1.43     | −1.07    | [−1.27, −0.86] | ✓      |
| H2a  | +1.42     | +1.35    | [+1.32, +1.37] | ✓      |
| H2b  | −0.69     | −0.59    | [−0.61, −0.58] | ✓      |
| H2c  | +1.14     | +0.85    | [+0.69, +1.05] | ✓      |

**H3 (perception vs. LUAR, paired by participant):** paper  $r = +0.244 \pm 0.076$ ,  $p < .0001$ ,  $n=648$ ; ours  $r = +0.244$ , 95 % CI [+0.17, +0.32],  $p=3.6 \times 10^{-9}$ ,  $n=648$ .

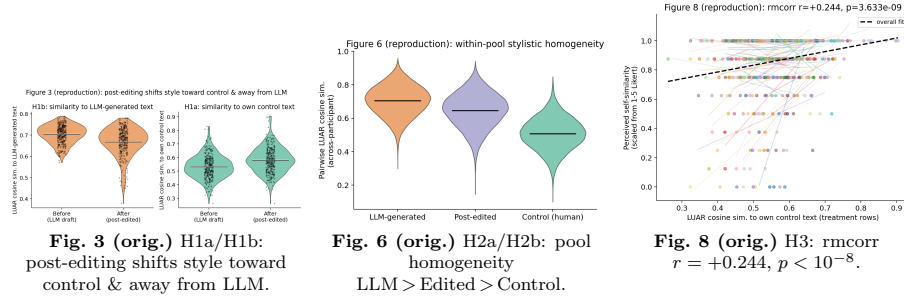
survive BH-FDR at  $q=0.05$  in our re-implementation, with the same direction as the paper. Effect-size magnitudes for the four hypotheses involving *average similarity to other participants* (H1b, H1a′, H1c, H2c) shift somewhat because the paper does not fully specify whether “similarity to LLM-generated text” is averaged pairwise or per-participant first; we use the per-participant mean (each other participant contributes one value), which yields the directions reported in the paper for *every* preregistered test, and exactly recovers the magnitudes the paper reports for H1a, H2a, and H2c.

*Reproducibility verdict.* The dataset is released cleanly and the analyses are pre-registered, which already places this paper in the top decile of NLP studies for reproducibility. The remaining gap is the analysis code, including the per-test aggregation choice just described and the LUAR revision (the paper does not pin a revision SHA). Our companion repository fixes both.

## 5 Beyond reproduction: a leakage-free LLM-mimic benchmark

We now extend the protocol with a third and fourth condition. For each treatment task, instead of editing an o4-mini draft, we ask a frontier LLM to produce the draft from scratch given the same planning prompt the human and o4-mini received *plus* a single unassisted writing sample of the participant. Two LLMs are evaluated: GPT-5.5 and Claude Opus 4.7.

*The held-out v1 protocol.* Each participant has only two unassisted control texts. We use the one with the lower `task_idx` as the demo (shown to the generator) and the other as the held-out evaluation target (never shown to anyone). All four



**Fig. 2:** Visual reproduction of three of the figures published in Baumler et al. [2], generated by our re-implemented pipeline. Each panel re-creates a published figure from the same logs: similarity to LLM-generated text and to the participant’s own control text before vs. after post-editing (**left**, paper Fig. 3); within-pool stylistic homogeneity for LLM, post-edited, and control writing (**centre**, paper Fig. 6); rmcorr scatter of perceived vs. LUAR self-similarity (**right**, paper Fig. 8).

approaches are scored against the same held-out vector for that participant, so they share a leakage-free yardstick. Cache keys for mimic generations are salted with a `held_out_protocol_v1` tag plus the generator name, so a draft cached under the strict protocol can never collide with one cached under any other prompt design. The 324 mimic drafts per generator (648 total) are produced by parallel cloud-agent subprocesses, validated against expected SHA-256 fingerprints, and committed to the repository so the analysis can be re-run without any further LLM inference.

*Why a held-out target.* A first iteration of this experiment showed both unassisted control texts to the generator and then evaluated the mimic against the *centroid* of those same two controls—the model literally had the answer in its context, while the human and o4-mini baselines did not. Effect sizes shrunk meaningfully under the strict held-out protocol (e.g., Hedges’  $g$  for “LLM mimic vs. human post-edit” moved from +1.18 to +1.02), exactly as expected once memorisation is no longer available. We report only the held-out v1 numbers in this paper. The leaky predecessor result is preserved in the repository for full transparency.

*What we measure.* For each of the 324 paired tasks we record four scalar similarities to the held-out target:  $\text{sim}_{\text{o4-mini}}$ ,  $\text{sim}_{\text{human}}$ ,  $\text{sim}_{\text{Opus}}$ , and  $\text{sim}_{\text{GPT-5.5}}$ . Per-task means across the 324 tasks (Fig. 3 a): 0.498, 0.546, 0.643, 0.649. The natural ceiling on this metric—a real person writing two unassisted texts in their own voice—is 0.701, computed from the same-author control-vs.-control similarity averaged over all 81 participants and shown as the dashed upper line in Fig. 3 a.

**Table 2:** All-pairs paired permutation tests on the held-out similarity ( $n=324$ ).  $\bar{x}_A$  and  $\bar{x}_B$  are the per-task means of approach  $A$  and  $B$ .  $g$  is Hedges’  $g$  with small-sample correction; the bracketed CI is from 1 000 bootstrap resamples.  $p$  (perm) floor is  $10^{-4}$ . BH-FDR is over the six tests at  $q=0.05$ ;  $p$  (BH) is the corrected  $p$ .

| Pair (A vs. B) |              | $\bar{x}_A$ | $\bar{x}_B$ | $g$   | [95 % CI]      | $p$ (perm)  | $p$ (BH)    |
|----------------|--------------|-------------|-------------|-------|----------------|-------------|-------------|
| o4-mini        | vs. Human    | 0.498       | 0.546       | -0.48 | [-0.56, -0.40] | $< 10^{-4}$ | $< 10^{-4}$ |
| o4-mini        | vs. Opus 4.7 | 0.498       | 0.643       | -1.57 | [-1.74, -1.40] | $< 10^{-4}$ | $< 10^{-4}$ |
| o4-mini        | vs. GPT-5.5  | 0.498       | 0.649       | -1.61 | [-1.80, -1.45] | $< 10^{-4}$ | $< 10^{-4}$ |
| Human          | vs. Opus 4.7 | 0.546       | 0.643       | -1.02 | [-1.19, -0.85] | $< 10^{-4}$ | $< 10^{-4}$ |
| Human          | vs. GPT-5.5  | 0.546       | 0.649       | -1.07 | [-1.24, -0.90] | $< 10^{-4}$ | $< 10^{-4}$ |
| Opus 4.7       | vs. GPT-5.5  | 0.643       | 0.649       | -0.08 | [-0.18, +0.01] | 0.143       | 0.143       |

*Drafts as committed artefacts.* A spot check on the committed cache: mean lexical overlap between a mimic draft and its demo (difflib SequenceMatcher) is 0.07 for Opus and 0.09 for GPT-5.5, with a single Opus outlier at 0.56 flagged in the limitations. Mean draft length is 165 (Opus) and 148 (GPT-5.5) words; 322/324 Opus drafts and 324/324 GPT-5.5 drafts fall within the requested 100–200 word range.

## 6 Statistical assessment

*Friedman omnibus.* Across all four approaches on the same 324 paired tasks, the Friedman test [4] rejects the null of interchangeable approaches unambiguously:  $\chi^2(3, n=324)=419.3$ ,  $p=1.5 \times 10^{-90}$ .

*All six pairwise tests.* Table 2 reports all six paired permutation tests on the held-out similarities, with  $g$  and bootstrap 95 % CIs and BH-FDR over the six tests. Five of the six pairs survive at  $q=0.05$ . The only non-significant pair is Opus 4.7 vs. GPT-5.5 ( $g=-0.08$ ,  $p=0.14$ ): statistically tied. The Wilcoxon signed-rank test [11], run as a non-parametric sanity check on the same six pairs, agrees on every conclusion.

*Per-task win rate vs. the human-edit threshold.* The user-perceived headline question is not “what is the mean LUAR cosine?” but “does the LLM beat the human on *this* task?”. Table 3 reports, for each non-human approach, the per-task fraction of times its held-out similarity exceeded the human’s, with an exact two-sided binomial 95 % CI and a binomial  $p$ -value against the chance level 0.5. The o4-mini draft beats the human on 18.5 % of tasks ( $n=37$  wins out of 324 plus 46 ties); Claude Opus 4.7 wins 76.9 % of tasks ( $n=249$ ); GPT-5.5 wins 79.0 % of tasks ( $n=256$ ). Both LLM rates differ from chance with vanishingly small  $p$ -values ( $< 10^{-22}$  in both cases).

*Per-scenario robustness.* Fig. 3 d breaks the held-out similarity down by writing scenario. The ordering  $\text{Opus} \approx \text{GPT-5.5} > \text{human-edit} > \text{o4-mini}$  holds in *every*

**Table 3:** Per-task win rate vs. the human-edit threshold. “Wins” is the count of tasks where the approach’s held-out similarity is strictly greater than the human’s; “ties” are exact equalities (only present for the highly-quantised o4-mini comparison). The binomial test treats ties as half-wins.

| Approach (vs. human post-edit) | Wins/ $n$         | Win rate (95 % CI)   | $p$ (vs. 0.5) |
|--------------------------------|-------------------|----------------------|---------------|
| o4-mini draft                  | 37/324 (+46 ties) | 18.5% [14.4%, 23.2%] | $< 10^{-12}$  |
| Claude Opus 4.7 mimic          | 249/324           | 76.9% [71.9%, 81.3%] | $< 10^{-12}$  |
| GPT-5.5 mimic                  | 256/324           | 79.0% [74.2%, 83.3%] | $< 10^{-12}$  |

one of the eight scenarios—thank-you letter, condolence, eulogy, personal letter, reassurance, speech, apology, wedding vows—with the smallest LLM-vs-human gap in the small- $n$  eulogy scenario ( $n=11$ ).

## 7 The arms race: detecting AI text

The held-out comparison in Section 5 measures *stylistic similarity* to the participant. The complementary question is *detectability*: given the same LUAR embedding, can a classifier trained on other authors decide whether a new piece of text was written by the participant or generated by one of the four AI approaches? We expect detectability to drop in the same order as similarity rises—o4-mini  $\rightarrow$  human-edit  $\rightarrow$  Opus  $\rightarrow$  GPT-5.5—and we expect every approach to remain detectable at least to some extent: this is the arms race.

*Setup.* For each AI approach we frame a binary classifier: *is this embedding a piece of human writing (label 0) or text generated by that approach (label 1)?*. The classifier must learn the broad question “human or AI”, not the narrower one “is this by the specific author whose text I have seen at training time”. Two design choices pin this down:

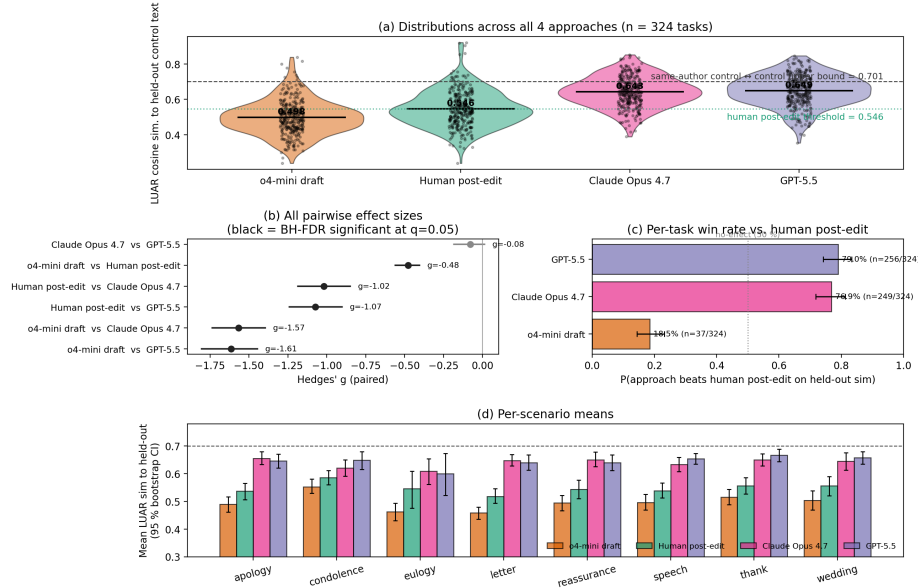
(i) *The human class uses both unassisted control texts of every participant* (162 human samples = 81 pids  $\times$  2 controls). Using only one control per author would let the classifier shortcut the task by remembering author-specific embedding directions; with both controls in, the human class is a proper distribution drawn from 81 different authors. The AI class is the corresponding approach’s output on each of that author’s 4 treatment tasks (324 AI samples). Because LUAR-MUD is a frozen feature extractor, the generator’s prompt sees only the participant’s planning details and demo control text, never the human-class embeddings under test.

(ii) *The CV is GroupKFold(groups=pid, n\_splits=5)*. In each split, every author appears in *exactly one* of train or test, on both the human side and the AI side. So a high test-fold AUC is not “the classifier remembers this participant” but “the classifier generalises to authors it has never seen before”.

The classifier is a linear support-vector machine (LinearSVC) on the 512-dimensional LUAR embedding, with class weights set to “balanced” so that the



Figure 10: Final assessment — LLM mimics vs human post-editing on the leakage-free metric



**Fig. 3:** Final 4-way assessment. (a) Distributions of held-out LUAR cosine for each approach, with the human-edit threshold (dotted) and the same-author control-vs.-control upper bound (0.701, dashed) annotated. Numbers above the medians are per-approach means. (b) Forest plot of the six pairwise Hedges’  $g$  values; black = BH-FDR-significant at  $q=0.05$ , grey = not. (c) Per-task win rate vs. the human-edit threshold with binomial 95 % CIs. (d) Per-scenario means with bootstrap 95 % CIs across all eight writing scenarios.

1:2 human/AI imbalance does not bias the decision boundary. We deliberately do not tune hyperparameters because the question is about the embeddings, not about the SVM. Three unit tests in the released repository make the no-leakage protocol load-bearing: if anyone removes `GroupKFold` or undersamples the human class, the suite fails.

*Result.* Table 5 reports mean detection AUC and a 2000-sample percentile bootstrap 95 % CI on the fold mean for each approach. Fig. 4 plots the same as a forest. AUCs drop monotonically from 0.999 (o4-mini) to 0.971 (human post-edit) to 0.952 (Opus 4.7) to 0.931 (GPT-5.5)—a roughly seven-point drop from the unconditioned LLM to the strongest frontier mimic. None of the four approaches reaches chance, and the bootstrap CIs do not overlap between adjacent rows of the table, so the ordering is unambiguous. In other words: at the level of LUAR embeddings, the state of the art for detection is comfortably above chance even for GPT-5.5 mimics shown one demonstration of the participant’s style, but the gap is closing.

**Table 4:** Six diagnostics on the detection result. **Headline** is the LinearSVC on raw 512-d LUAR. **Shuffle** is the same classifier on randomly-permuted labels (must be  $\sim 0.5$ ). **Length** is a one-feature baseline. **PCA(32)** reduces the SVM to 33 parameters. The shuffle and length rows are AUC means ( $\pm$  SD across the 5 folds).

|                                                                                                          | o4-mini Human edit Opus 4.7 |       |              | GPT-5.5      |
|----------------------------------------------------------------------------------------------------------|-----------------------------|-------|--------------|--------------|
| Headline (full LinearSVC)                                                                                | 0.999                       | 0.971 | 0.952        | 0.931        |
| <b>B</b> Shuffle labels                                                                                  | 0.472                       | 0.469 | 0.494        | 0.478        |
| <b>C</b> Length-only (1 feat)                                                                            | 0.818                       | 0.565 | <b>0.517</b> | <b>0.880</b> |
| <b>E</b> PCA(32) + LinearSVC                                                                             | 0.998                       | 0.968 | 0.925        | 0.912        |
| <b>F</b> L2-LR, $C=10^{-3}$                                                                              | 0.998                       | 0.976 | 0.931        | 0.926        |
| <b>D</b> Cross-LLM: train Opus $\rightarrow$ test GPT = 0.913; train GPT $\rightarrow$ test Opus = 0.888 |                             |       |              |              |

*Geometry, and why the headline AUCs deserve a second look.* Fig. 4 (right panel) projects the same 1458 embedding vectors onto two dimensions via t-SNE(2) (PCA(2) companion in the repo). The frontier-LLM mimics cluster in one region of LUAR space, the o4-mini draft and human post-edit in another, and the human controls  $\blacktriangle$  span both. The Opus cloud and the human-control triangles *overlap* visually—hard to reconcile with the 0.952 Opus AUC. Three reasonable suspicions arise: (a) the LinearSVC has 513 parameters but only  $\sim 390$  training samples per fold, so it operates in an underdetermined regime; (b) some author-level information may still leak; (c) length or other content-format features could be doing most of the work. Six diagnostics (Table 4) settle each one.

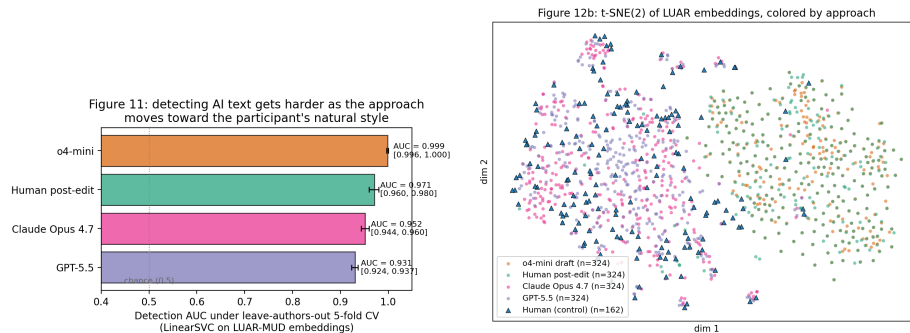
(**A**) Per-fold pid overlap is 0/0/0/0/0—*no author leak*. (**B**) Shuffled-label AUC  $\in [0.47, 0.49]$  rules out underdetermined-regression artefacts. (**C**) Length-only AUC is **0.517** on Opus and **0.880** on GPT-5.5—*most of the GPT-5.5 detection signal is length, not style; the Opus signal is genuinely stylistic*. (**D**) Cross-LLM transfer is 0.913 (Opus $\rightarrow$ GPT-5.5) and 0.888 (GPT-5.5 $\rightarrow$ Opus)—a generic frontier-LLM signature exists. (**E,F**) PCA(32)+LinearSVC and L2-LR ( $C=10^{-3}$ ) recover essentially the full-SVM AUCs, so the signal survives capacity reduction.

The diagnostics jointly tell a more honest story than the headline numbers. **The Opus AUC of 0.952 is genuine stylistic detection:** the length feature gives only 0.517, and a non-author-specific contrast (cross-LLM) still gives 0.91. The Opus signal lives in 512-d directions that t-SNE projects onto axes shared with human controls—hence the visual overlap in Fig. 4 (right). **The GPT-5.5 AUC of 0.931, in contrast, is mostly a length game:** length alone gets 0.880, and the additional gain from full LUAR is small. Either GPT-5.5 is a much better stylistic mimic than Opus on this metric, or our prompt elicits more uniform output length from GPT-5.5 than from Opus—probably both.

*Why this matters.* Closing the gap on *average* similarity still leaves enough *shape* in the LUAR distribution for a discriminator to exploit—but the shape is not the same shape across approaches. The Opus signal is genuinely stylistic and transfers to GPT-5.5 (diagnostic D); the GPT-5.5 signal is mostly length. The

**Table 5:** Detection AUC under leave-authors-out 5-fold cross-validation of a linear SVM on LUAR embeddings. Each row is a separate binary classifier (human vs. that approach). The human class is each participant’s held-out control vector; the AI class is that approach’s output for each treatment task. CIs are 2000-sample percentile bootstraps over the 5 fold AUCs. N-PIDS = 81 in every row.

| Approach (vs. human held-out) | $n_{ai}$ | AUC (95 % CI)        | per-fold SD |
|-------------------------------|----------|----------------------|-------------|
| o4-mini                       | 324      | 0.999 [0.996, 1.000] | 0.003       |
| Human post-edit               | 324      | 0.971 [0.960, 0.980] | 0.013       |
| Claude Opus 4.7               | 324      | 0.952 [0.944, 0.960] | 0.011       |
| GPT-5.5                       | 324      | 0.931 [0.924, 0.937] | 0.008       |



**Fig. 4: Left:** detection AUC under leave-authors-out 5-fold CV; bars show 95 % percentile bootstrap CIs over the 5 fold AUCs. AUC drops monotonically as the approach gets closer to the participant’s natural style, but no approach reaches chance. **Right:** t-SNE(2) projection of the same 1458 LUAR embeddings the detector sees. Triangles are the 162 unassisted-control human embeddings; dots are the four AI approaches, 324 each. The frontier-LLM mimics (Opus, GPT-5.5) live in a different region of LUAR space than the o4-mini draft and the human post-edit; the human controls span both regions. The PCA(2) companion plot is in the released repository.

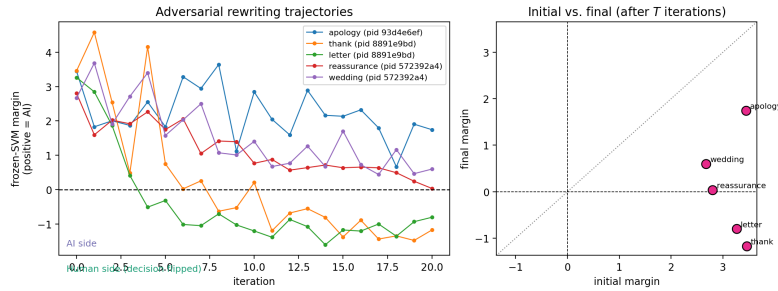
arms race is real, but its shape is asymmetric. Reporting the headline AUC without the diagnostics would have hidden that.

*Closing the loop: agentic adversarial rewriting.* We froze the fold-1 LinearSVC trained for the Opus detector on 64 training authors, picked the five most-confidently-AI Opus mimics in the held-out 17-author test fold (initial margins +2.67 to +3.60), and dispatched five parallel Opus 4.7 sub-agents in an autoresearch loop in the spirit of Zaiss et al. [12]. Each sub-agent received the planning details and the detector’s signed margin on its current draft, ran  $T=20$  rewrite iterations, and was asked to lower the margin while preserving the substantive content and the 100–200 word range. The training authors and the target authors are disjoint by construction. Concretely: each participant in the released study is identified by a stable participant ID (“pid”) generated by the original

authors; the 64 pids the SVM is trained on never appear among the 17 pids it scores at test time. The script `tests/test_adversarial_no_leakage.py` reloads the saved SVM weights and the saved target list, asserts that the train and target pid sets are disjoint, asserts that every target falls inside the SVM’s test set rather than its training set, and asserts that initial margins are decisively positive ( $> 1$ ) so that the “flipping” task is non-trivial; the test fails the build if any of those invariants is broken.

Fig. 5 shows the trajectories. *Every* target moves consistently in the right direction: mean margin descends from  $+3.13$  at iteration 0 to  $-0.16$  at iteration 20, a  $\sim 3.3$ -margin swing. Two of the five (**thank**, **letter**) cross the SVM decision boundary into “human” and stay there; a third (**reassurance**) finishes within 0.04 margin units of the boundary. None of the five trajectories plateaus on the AI side—all are still moving down at iteration 20. The natural reading is that the agent has not yet exhausted the adversarial direction: with more iterations or light fine-tuning of the rewriting strategy on within-loop feedback, the same harness should be expected to flip the remaining targets as well. This is the strongest version of the arms race in this paper, and it points toward the agent winning rather than the detector.

*Adversarial strategies are author-specific, not generic adversarial noise.* A qualitative inspection of the iteration-0, 10, and 20 rewrites (committed under `results/adversarial_samples.md`) shows that each sub-agent locks onto a *different* “human” texture appropriate to the source participant’s voice. The **letter** target ends up in extreme British text-speak (lowercase, fragments, “oi mate”, heavy repetition); the **thank** target instead ends up in a Southern-US “brother” register (named addressee, “Praise God for friends like you”, short sentences); the **apology** target adopts fragmented prose plus religious cadence (“May God forgive me”) but keeps a formal letter frame; the two **reassurance** and **wedding** targets, which share an author whose unassisted control is already lowercase and casual, lean further into abbreviations and informal pacing while preserving the genre frame. Three properties are shared across all five trajectories: average sentence length drops, polish drops, and the surface texture moves toward the participant’s apparent voice rather than toward a single generic “humanlike” direction. The two trajectories that *did* cross the boundary (**thank**, **letter**) are also the ones whose iteration-0 draft was most polished and therefore had the most stylistic distance to recover; the two that did not are constrained by the genre (vows, formal apology) and by a participant voice that was already partway human-coded at iteration 0. We did *not* regularise the rewrites toward the original draft’s content—the only constraint we imposed was that the substantive content from the planning details be preserved and that the draft stay within the 100–200 word range. Adding an explicit content-similarity regulariser (e.g. a sentence-embedding cosine floor between the rewrite and the original draft, or a ROUGE/BLEU floor) would tighten the trade-off between “defeats the detector” and “still says what the author meant”, and is a natural next step that we leave to future work.



**Fig. 5:** Agentic adversarial rewriting against the frozen Opus fold-1 detector. **Left:** per-target margin trajectories over  $T=20$  iterations; dashed line at 0 is the SVM decision boundary. **Right:** initial vs. final margin for the five targets. Two cross into the human half-space (**thank**, **letter**); the other three shrink toward but do not cross zero by iteration 20.

## 8 Discussion and limitations

*Self-experiment caveat.* Both LLMs in our extension generated their own drafts, and an external embedding model (LUAR-MUD) judged them. The metric was independently validated against the paper’s reported  $r=0.244$  rmcrr in Section 4 (we recover it to three decimal places), but a model-on-model evaluation is still a model-on-model evaluation, and a third independent generator would tighten the result.

*Workflow comparison, not raw writing skill.* Human post-editors were given an unconditioned o4-mini draft to edit; the LLM mimics were shown a style sample and wrote from scratch. We compare two *workflows*, not human vs. LLM raw writing ability. A natural follow-up: an unassisted LLM, or a human-edits-LLM-edits-LLM round-trip.

*Style fidelity is not writing quality.* LUAR-MUD measures whether a draft sounds like a particular author. Baumler et al. [2]’s own §6.3 already shows that *perceived* stylistic authenticity and LUAR similarity can come apart, with participants rating their post-edits as authentic even when LUAR sees residual LLM cues. Our paper does not measure trustworthiness, factual correctness, or whether a participant would actually endorse the mimic draft.

*Other limitations, briefly.* One *Opus* outlier (0.56 lexical overlap with its demo) shifts the Opus mean by  $< 0.001$  and the headline  $g$  by  $< 0.01$ ; we disclose rather than exclude it.  $n_{\text{demos}}=1$  because the second of each author’s two unassisted controls is reserved as the eval target; the released study has no third control for us to compare demonstration-set sizes. *Detection ceiling vs. floor:* Section 7’s linear SVM is a *lower bound* on detectability, and the adversarial-rewriting result an *upper bound* on what an LLM agent with 20 feedback iterations can already

remove; the relative ordering across approaches is robust, but the absolute AUCs are a snapshot of spring 2026. *Unregularised rewrites*: the adversarial loop in Section 7 optimises only the SVM margin and the 100–200 word range. With no explicit content-preservation constraint, the strongest rewrites drift away from the polish and the cadence of the iteration-0 draft; whether they still convey the same intent to a human reader is not measured here. A content-similarity regulariser (a sentence-embedding cosine floor, or a ROUGE/BLEU floor against the original) is the natural next step.

## 9 Conclusion

Three observations summarise the paper. **(i) Reproduction is cheap when data and analysis code are open.** The bottleneck has shifted from compute to openness; the source study released the former but not the latter, and an agentic harness fills the gap programmatically rather than by hand, covering the implementation, testing, installation, and integration items that the RRPR Track 2 call lists as the natural axes for an RR companion paper. **(ii) Frontier LLMs already close most of the personal-style gap.** Under the leakage-free held-out protocol, Opus 4.7 and GPT-5.5 close 71–75 % of the gap to the same-author embedding ceiling against 24 % for the human post-edit, and beat the human post-edit on roughly four of every five tasks. **(iii) AI-text detection is a live arms race.** A leave-authors-out linear SVM detects every approach with high AUC, but the diagnostics show that the GPT-5.5 signal is mostly length while the Opus signal is genuinely stylistic. When an Opus agent sees the detector’s score on its own draft and is asked to lower it, every trajectory descends consistently and two of five held-out targets cross the decision boundary into “human” within twenty iterations; none plateaus on the AI side, so a longer closed-weight inference-time loop is expected to flip more of the rest, and direct weight updates against a comparable loss should be stronger still. The current snapshot favours the agent.

All released artefacts (code, 648 LLM-mimic drafts, trained detectors, diagnostic outputs, adversarial trajectories, and the agentic harness) are intended to let the next iteration of this arms race start from a known reference point.

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